

Problem 1

KK 2.1: A 5kg mass is influenced by the force $\mathbf{F} = (4t^2\hat{\mathbf{i}} - 3t\hat{\mathbf{j}})\text{N}$.

(a)

From Newton's second law, we have $\mathbf{a} = \mathbf{F}/m$. We integrate once to find the velocity:

$$\mathbf{v} = 0 + \frac{1}{5} \int_0^t (4t'^2\hat{\mathbf{i}} - 3t'\hat{\mathbf{j}}) dt' = \frac{1}{5} \left(\frac{4}{3}t^3\hat{\mathbf{i}} - \frac{3}{2}t^2\hat{\mathbf{j}} \right) \text{ m/s} \quad .$$

(b)

Integrating once more, we find the position

$$\mathbf{r} = 0 + \frac{1}{5} \int_0^t \left(\frac{4}{3}t'^3\hat{\mathbf{i}} - \frac{3}{2}t'^2\hat{\mathbf{j}} \right) dt' = \frac{1}{5} \left(\frac{1}{3}t^4\hat{\mathbf{i}} - \frac{1}{2}t^3\hat{\mathbf{j}} \right) \text{ m} \quad .$$

(c)

We may now compute $\mathbf{r} \times \mathbf{v}$ and find

$$\mathbf{r} \times \mathbf{v} = \frac{1}{25} \left(-\frac{4}{3}t^3\frac{1}{2}t^3 + \frac{3}{2}t^2\frac{1}{3}t^4 \right) \hat{\mathbf{k}} = -\frac{t^6}{150} \hat{\mathbf{k}} \text{ m}^2/\text{s}$$

Problem 2

KK 2.16 We are given the following three constants of nature

$$h = 6.6 \cdot 10^{-34} \text{ Js} = 6.6 \cdot 10^{-34} \text{ kg m}^2 \text{ s}^{-1}$$

$$G = 6.7 \cdot 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$$

$$c = 3.0 \cdot 10^8 \text{ m s}^{-1}$$

To form new quantities that do not include kg, it is clear we need to multiply h and G to get $[hG] = \text{m}^5\text{s}^{-3}$. We can then cancel the s^{-3} by dividing by c^3 such that $[hG/c^3] = \text{m}^2$. Taking the square root, we find the length

$$L_p = \sqrt{\frac{hG}{c^3}} = \sqrt{\frac{6.6 \cdot 10^{-34} \cdot 6.7 \cdot 10^{-11}}{(3.0 \cdot 10^8)^3}} = 4.0 \cdot 10^{-35} \text{ m} \quad .$$

On the other hand, we similarly notice that $[hG/c^5] = \text{s}^2$ and the Planck time becomes

$$T_p = \sqrt{\frac{hG}{c^5}} = \sqrt{\frac{6.6 \cdot 10^{-34} \cdot 6.7 \cdot 10^{-11}}{(3.0 \cdot 10^8)^5}} = 1.4 \cdot 10^{-43} \text{ s} \quad .$$

Finally, to obtain a quantity with the unit of kg, we notice that $[h/G] = \text{kg}^2 \text{ s/m}$, such that the following form has the correct dimension

$$M_p = \sqrt{\frac{hc}{G}} = \sqrt{\frac{6.6 \cdot 10^{-34} \cdot 3.0 \cdot 10^8}{6.7 \cdot 10^{-11}}} = 5.4 \cdot 10^{-8} \text{ kg} \quad .$$

Problem 3

KK 2.6 The force $\mathbf{F}_w$ of the cone-wall is normal to the surface. The centrifugal

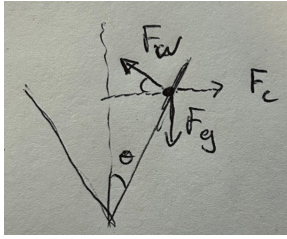


Figure 1: Force diagram for the cone in Problem 3.

force $|\mathbf{F}_c| = mv_0^2/r$ is directed horizontally outwards, whereas the gravitational force $|\mathbf{F}_g| = mg$ acts vertically downward.

In the polar coordinate system, the mass does not move in r -direction or in the up-down (z)-direction, such that horizontal force balance requires $\Sigma F_r = 0$, while vertical force balance implies $\Sigma F_z = 0$.

Thus, vertical force balance yields

$$F_w \sin \theta = F_g \quad ,$$

while horizontal force balance yields

$$F_w \cos \theta = F_c \quad .$$

By taking the ratio, we find

$$\frac{F_w \sin \theta}{F_w \cos \theta} = \tan \theta = \frac{mg}{mv_0^2/r} \quad .$$

Solving for r yields

$$r = \frac{v_0^2 \tan \theta}{g} \quad .$$

Problem 4

KK 2.11: .

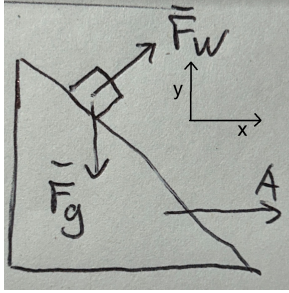


Figure 2: Force diagram for the block and wedge in Problem 4.

The interesting part of this problem is the constraint that the sloped surface imposes on the block. If the y -coordinate of the block is fixed, then it is clear that the block will accelerate with the 45° wedge in the x -direction, such that $a_x = A$. Meanwhile, if the block moves upwards Δy in the y -direction, it will move a distance $\Delta x = -\Delta y$ in the x -direction. It follows that the a_x and a_y accelerations of the block must be related through the following constraint:

$$a_x = A - a_y \quad .$$

Applying Newton's second law in the x and y -directions, we get

$$ma_x = \frac{|\mathbf{F}_W|}{\sqrt{2}} \quad , \quad ma_y = \frac{|\mathbf{F}_W|}{\sqrt{2}} - mg \quad .$$

Dividing through by m and inserting the resulting expressions for a_x and a_y into the constraint, we obtain

$$\frac{|\mathbf{F}_W|}{m\sqrt{2}} = A - \frac{|\mathbf{F}_W|}{m\sqrt{2}} + g \quad .$$

This equation is readily solved for $|\mathbf{F}_W|$, yielding

$$|\mathbf{F}_W| = \frac{mg + mA}{\sqrt{2}} \quad .$$

Inserting this into the previous expressions for a_x and a_y , we obtain the desired answers

$$a_x = \frac{g + A}{2} \quad , \quad a_y = \frac{A - g}{2} \quad .$$

Problem 5

KK 2.7:

(a)

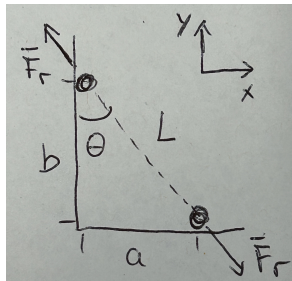


Figure 3: Force diagram for the rod in Problem 5.

To find a constraint between a_x and a_y , we notice from the Pythagorean theorem that $x^2 + y^2 = L^2$. Differentiating once with respect to time yields $2x\dot{x} + 2y\dot{y} = 0$. Differentiating once more yields $2\dot{x}^2 + 2x\ddot{x} + 2\dot{y}^2 + 2y\ddot{y} = 0$. We may also write this as

$$v_x^2 + xa_x + v_y^2 + ya_y = 0 \quad .$$

(b)

Initially, we have $v_x = v_y = 0$, and from the drawing, it follows that $y = b$. It is also useful to introduce the length $a = \sqrt{L^2 - b^2}$, representing the x -coordinate for the lower right mass at $t = 0$. Furthermore, if we let θ be the angle indicated between the y -direction and the rod, then $\cos \theta = b/L$ and $\sin \theta = a/L$.

Applying the initial conditions to the constraint from (a) yields

$$aa_x = -ba_y \quad .$$

Additional expressions involving a_x and a_y can be obtained by considering force balance in the x and the y -directions:

$$F_r \frac{a}{L} = ma_x \quad , \quad -mg + F_r \frac{b}{L} = ma_y \quad .$$

After intersecting these into the constraint, we find

$$a \frac{F_r}{m} \frac{a}{L} = -b \left(-g + F_r \frac{b}{mL} \right) \quad ,$$

or

$$\frac{F_r}{m} \frac{a^2 + b^2}{L} = bg \quad .$$

But, because $a^2 + b^2 = L^2$, it follows that

$$F_r = \frac{b}{L} mg \quad .$$

Inserting this into the force balance equations above, we find the desired answers

$$a_x = \frac{ab}{L^2} g \quad , \quad a_y = \left(\frac{b^2}{L^2} - 1 \right) g \quad .$$

These may also be rewritten as

$$a_x = \sin \theta \cos \theta g \quad , \quad a_y = -\sin^2 \theta g \quad .$$